UAA College of Engineering
UNIVERSITY of ALASKA ANCHORAGE

Magnetostrictive Sensor for Detection of Internal Corrosion in Unpiggable Pipelines

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Background

Production of oil and gas in Alaska necessitates operation of large networks of pipelines and implementation of various corrosion control and mitigation measures. Even with these measures, incidents involving spillage of oil still occur.

Smaller pipelines (flow lines and facility piping), represent the most threatened assets. They have relatively small diameter and are rarely equipped with launching and retrieval infrastructure for Pipeline Inspection Gauges (PIGs) that are used for detection of defects on the pipe wall. Typically, these pipes are inspected by field crews using ultrasonic tools.

Figure 1: Preparation of a PIG device for launch.
https://www.pipingengineer.org/pipeline-pigging/

The industry is searching for a monitoring solution which will have the following characteristics:

(i) Safe in volatile environments,
(ii) Does not require alteration or have any interference with existing pipe structure,
(iii) Does not disturb operation of the pipeline during installation, operation, and replacement whenever necessary,
(iv) Cost-effective and being capable of being deployed over large distances.

Objective

The goal of this project is to investigate detection of pipe wall thinning and deterioration due to internal corrosion via monitoring of magnetic flux leakage by employing a fiber optic strain sensor attached to the substrate made of magnetostrictive material (Terfenol-D alloy).

Fiber Optic Sensors

- Can be deployed over long distances.
- Safe in volatile environments.
- Fiber Bragg Grating (FBG) sensor concept.

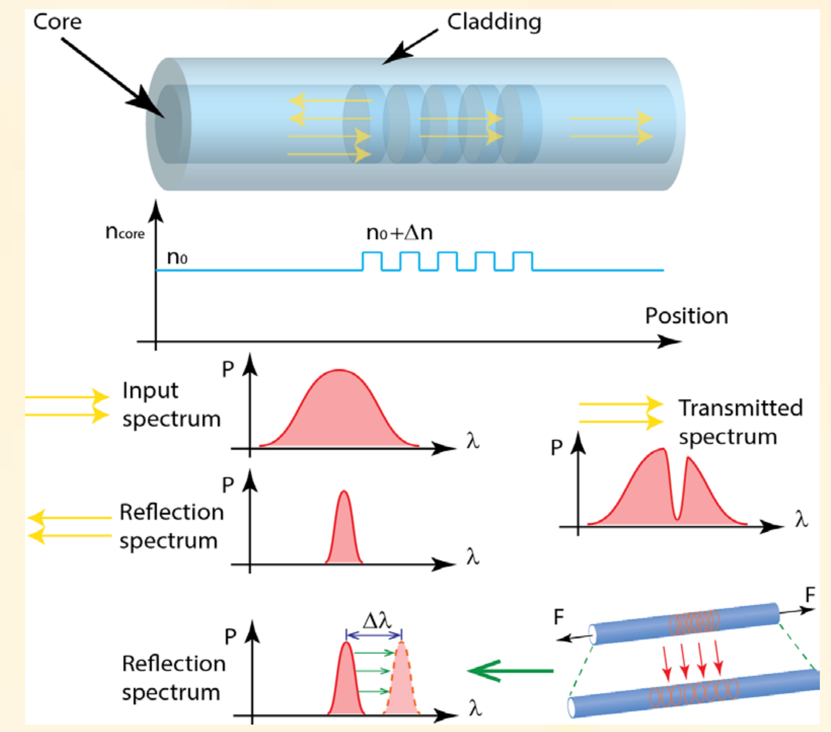

Figure 2: Detection of strain and temperature using a Fiber Bragg Grating sensor.

Magnetostrictive Sensor Concept

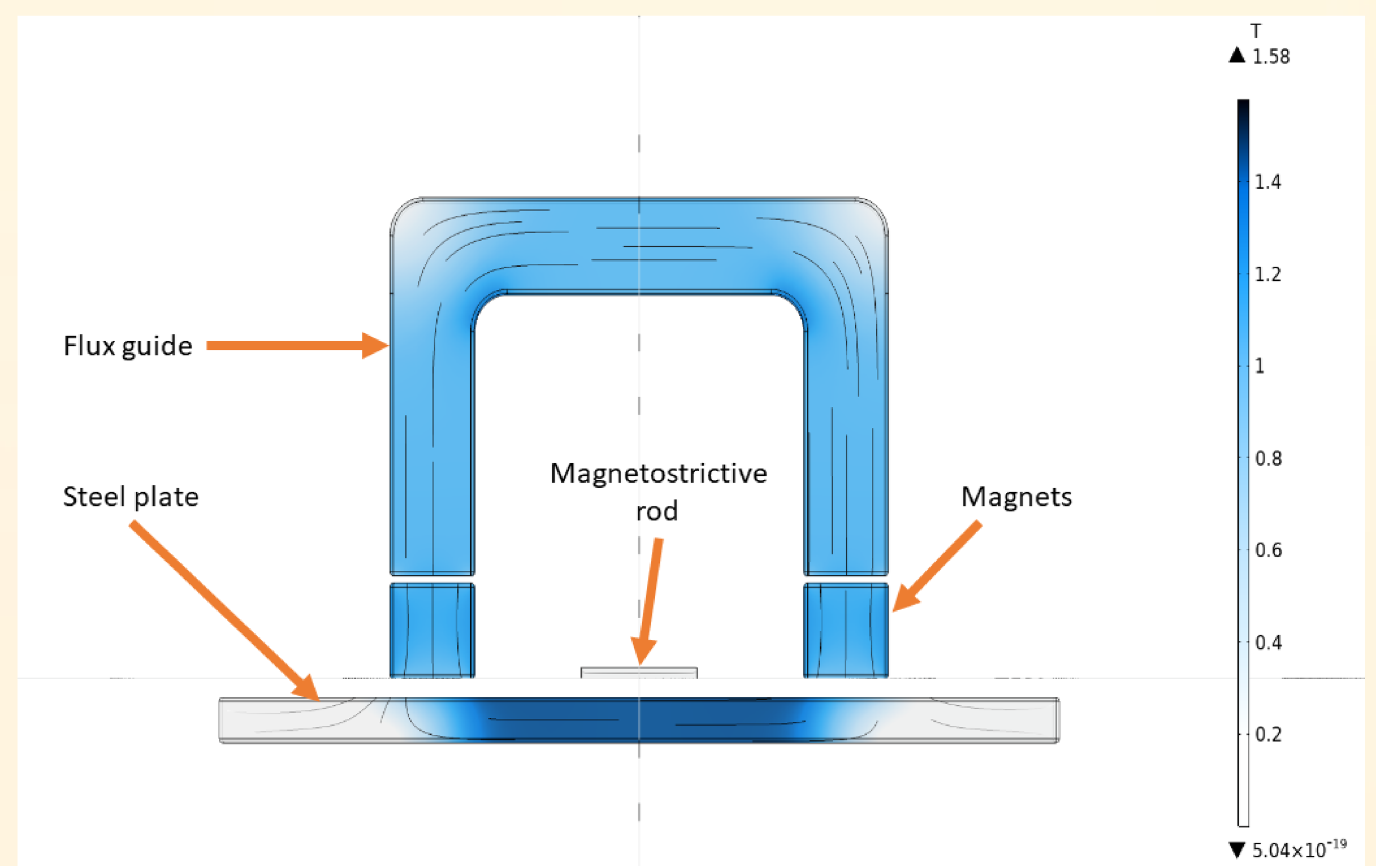

Figure 3: Flux distribution in the sensor model.

As the pipe wall gets thinner, magnetic flux leaks out and induces mechanical strain in the sensing rod made of Terfenol-D magnetostrictive alloy. Axial strain is measured via a fiber optic sensor glued on the rod. Terfenol-D rods with square cross-section are utilized in this work.

Compensation of thermal strain components is realized via using a rod of similar dimensions made of Inconel alloy.

- Non-magnetic, fully austenitic alloy
- Similar thermal properties with Terfenol-D

Experimental Setup and Results

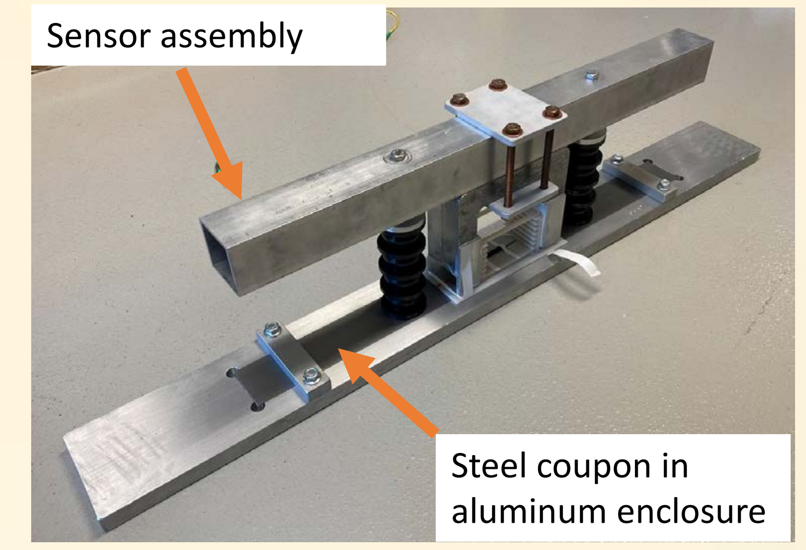

Figure 4: Sensor assembly in the lab.

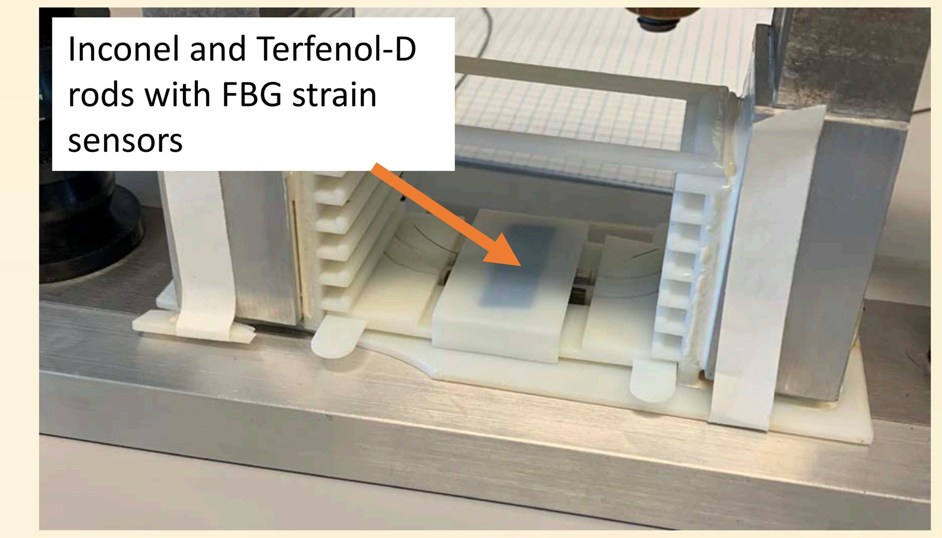

Figure 5: Close-up of the sensor assembly with the sensing tray carrying rods.

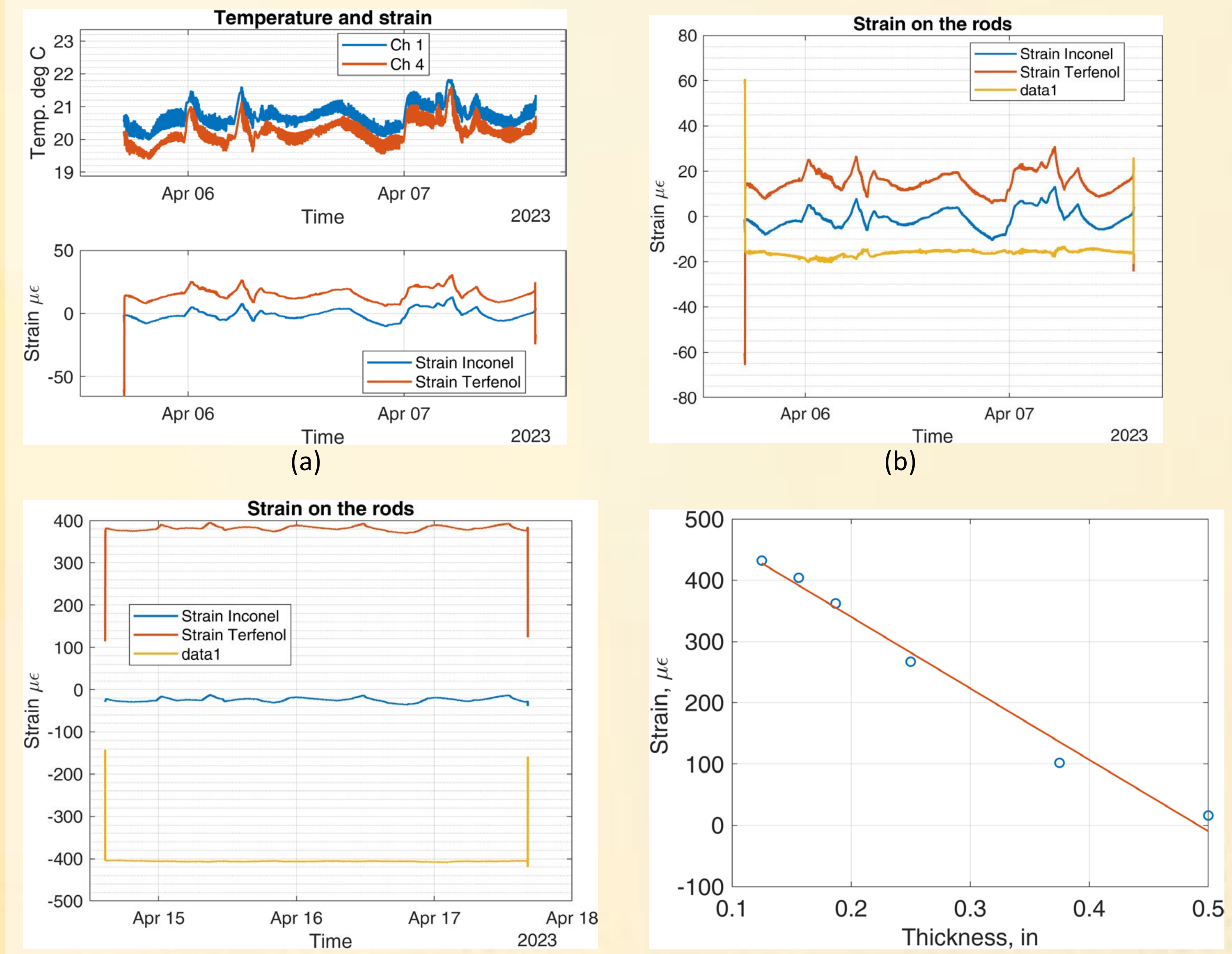

Figure 6: (a) Responses of 4x4 mm rods with 0.5 in. thick steel plate; (b) Responses plotted along with thermally compensated strain data, 0.5 in. thick steel plate; (c) Responses plotted along with thermally compensated strain data, 0.156 in. thick steel plate; (d) Average thermally compensated strain vs. steel thickness.

Conclusions and Future Work

- Strain changes linearly with reduction in steel wall thickness.
- Thermal strain compensation is implemented using the Inconel alloy.
- Patent application is currently in the works.

Acknowledgment

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